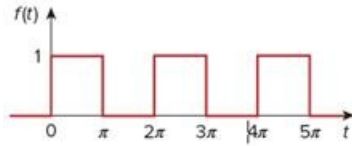


Problem

(a) Show that a periodic signal with exponential Fourier series $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$ has the Fourier transform $F(\omega) = \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$ where $\omega_0 = 2\pi/T$.

(b) Find the Fourier transform of the signal in Fig. 18.37.

Figure 18.37



Step-by-step solution

Step 1 of 2

$$\begin{aligned} \text{(A)} \quad F(C_n) &= C_n \delta(\omega) \\ F(C_n e^{jn\omega_0 t}) &= C_n \delta(\omega - n\omega_0) \\ F\left(\sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}\right) &= \sum_{n=-\infty}^{\infty} C_n \delta(\omega - n\omega_0) \end{aligned}$$

Step 2 of 2

$$\begin{aligned} \text{(B)} \quad T &= 2\pi \rightarrow \omega_0 = \frac{2\pi}{T} = 1 \\ C_n &= \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt = \frac{1}{2\pi} \left[\int_0^\pi 1 e^{-jnt} dt + 0 \right] \\ &= \frac{1}{2\pi} \left[\frac{-1}{jn} e^{-jnt} \right]_0^\pi = \frac{j}{2\pi n} (e^{-jn\pi} - 1) \\ \text{but} \quad e^{-jn\pi} &= \cos n\pi + j \sin n\pi \\ &= \cos n\pi = (-1)^n \\ C_n &= \frac{j}{2\pi n} [(-1)^n - 1] = \begin{cases} 0, & n = \text{even} \\ \frac{-j}{n\pi}, & n = \text{odd}; n \neq 0 \end{cases} \\ \text{for } n &= 0 \\ C_n &= \frac{1}{2\pi} \int_0^\pi dt = \frac{1}{2} \\ \text{Hence} \quad f(t) &= \frac{1}{2} - \sum_{\substack{n=-\infty \\ n \neq 0 \\ n = \text{odd}}}^{\infty} \frac{j}{n\pi} e^{jnt} \\ F(\omega) &= \frac{1}{2} \delta(\omega) - \sum_{\substack{n=-\infty \\ n \neq 0 \\ n = \text{odd}}}^{\infty} \frac{j}{n\pi} \delta(\omega - n) \end{aligned}$$